Brine Pretreat D_i TEA

Produced-Water Lithium Extraction

Rezaee DES/TOPO Li/Na bridge for PrOMMiS and IDAES integration

Chemical engineering technical review

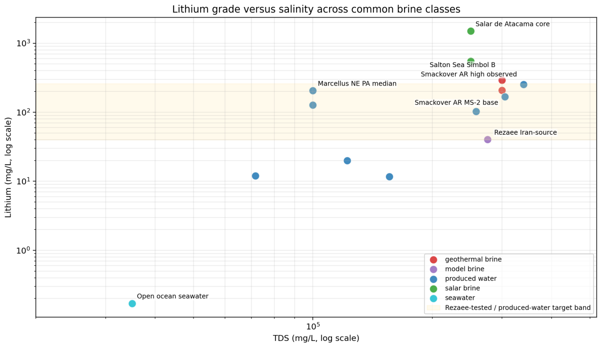
Smackover brine basis → divalent pretreatment → Rezaee DES/TOPO Li/Na bridge → PrOMMiS/IDAES handoff

May 8, 2026

Traceable Li/Na transfer variables are the bridge from produced-water chemistry to process economics.

CASE BASIS

Why Smackover Is The Flagship Feed



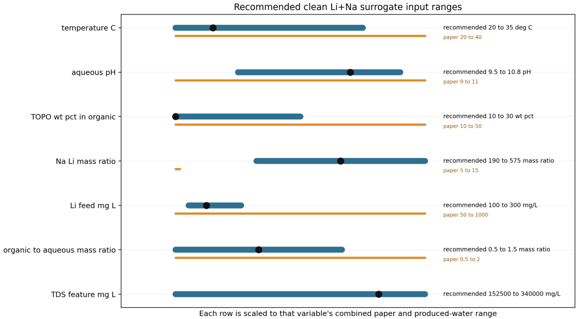
Engineering premise

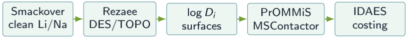
Southern Arkansas Smackover brine couples high lithium value with hypersaline electrolyte severity.

- Selected clean-feed basis: 168 mg/L Li and 64 100 mg/L Na.
- TDS is retained as a severity variable.
- Raw divalents are pretreatment context, not active solvent-model competitors.

Sources: USGS Smackover data release; local screening table.

Pretreatment Defines The Extraction Feed





Transfer variables, validity flags, and costing basis stay traceable

- The solvent model sees a clean Li/Na stream.
- Ca^{2+}, Mg^{2+}, Sr^{2+}, and Ba^{2+} are removed upstream.
- Produced-water severity is carried through TDS and Na/Li flags.

THERMODYNAMIC BRIDGE

Rezaee DES/TOPO Becomes A Transfer-Variable Bridge

THERMODYNAMIC SURROGATE CONTRACT

Model basis: calibrated_rezaee_section32_doe_basis_surface_distribution_surrogate

Caveat carried per row: outside original Rezaee Na/Li DOE domain

- Rezaee extraction responses are converted to Li and Na distribution coefficients.
- Calibrated $\log D_i$ surfaces generate the produced-water screening matrix.
- Model-basis and validity flags are carried into downstream process tables.

Sources: Rezaee et al. 2025 extraction data; Rezaee et al. 2026 parameter and reaction-equilibrium support.

Distribution Coefficients Are The Process Contract

$$D_i = \frac{E_i}{100 - E_i} \frac{1}{O/A}$$

$$S_{\mathrm{Li/Na}} = \frac{D_{\mathrm{Li}}}{D_{\mathrm{Na}}}$$

Nominal MS-2, $O/A = 1$

$E_{\mathrm{Li}} = 50.66\%,\ E_{\mathrm{Na}} = 5.67\%,$
$S_{\mathrm{Li/Na}} = 17.09.$

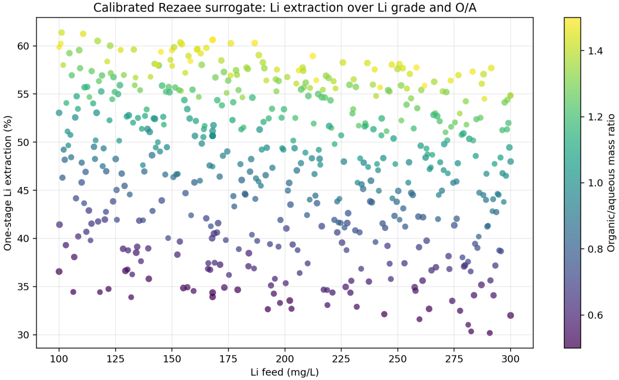
Nominal MS-2 transfer variables

$D_i = (E_i/(100 - E_i))/(O/A)$

E_{Li}	E_{Na}	$S_{Li/Na}$	D_{Li}	D_{Na}
50.66%	**5.67%**	**17.09**	**1.0267**	**0.0601**

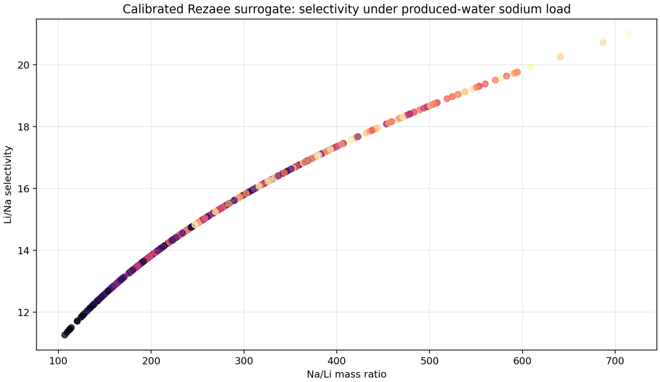
O/A = 1.0, Li = 168 mg/L, Na = 64,100 mg/L

Lithium Extraction Surface



- Current matrix spans about 30.2–61.4% one-stage Li extraction.
- Organic/aqueous ratio is the dominant operating lever.
- The result is a calibrated process-integration bridge, not a final direct reactive-LLE proof.

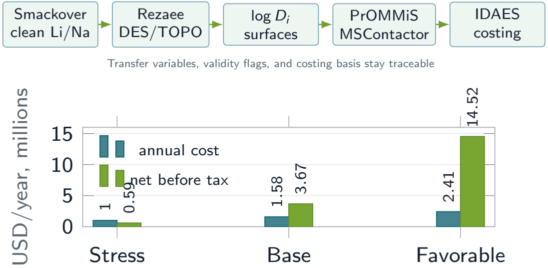
Sodium Load Controls The Validation Risk



- Selectivity remains favorable under produced-water sodium load.
- The MS-2 point is outside the original Rezaee Na/Li DOE domain.
- High-Na/Li validation is the most important next experimental check.

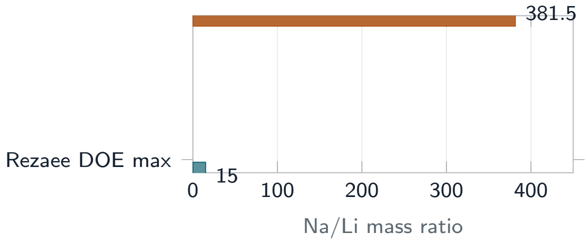
PROCESS INTEGRATION

PrOMMiS/IDAES Handoff



- Handoff rows carry D_i, selectivity, extraction, raffinate, and extract loading.
- Costing scenarios use the same feed-flow basis as the surrogate.
- Formal TEA still needs residence time, solvent loss, reagent demand, and product pricing.

Validation Boundary



Boundary condition

Every produced-water row is flagged as outside the original Rezaee Na/Li DOE training domain.

- Process screening is appropriate.
- Final physical acceptance requires high-Na validation.
- Direct reactive-LLE closure is not claimed in the main result.

The case study is ready for PrOMMiS/IDAES integration when D_i, $S_{\mathrm{Li/Na}}$, and validity flags remain traceable from the Rezaee basis.

Final Technical Message

1. Use Rezaee DES/TOPO as the active Li/Na case-study bridge.
2. Present Smackover as the produced-water source basis and pretreatment context.
3. Move only traceable transfer variables into PrOMMiS/IDAES.
4. Keep archived HBTA/TOPO, Gando, Jang, Yu, and Hubach material outside the main chain unless explicitly labeled as comparison or diagnostics.

Next rigorous step

Replace the algebraic process handoff with a full PrOMMiS MSContactor solve and an IDAES costing block once the final surrogate run interface is fixed.

References And Source Artifacts

- Rezaee et al. 2025 paper and supporting information: extraction response basis.
- Rezaee et al. 2026 paper and supplementary material: PC-SAFT/ePC-SAFT parameter and reactive-equilibrium evidence.
- U.S. Geological Survey southern Arkansas Smackover data release, DOI: `10.5066/P9QPRYZN`.
- Project artifacts: `analyses/rezaee_2026_pcsaft_epcsaft/` and `data/reference/produced_water/`.